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Introduction

The Surface Management System (SMS) **provides situational awareness with respect to airports, during take-off and landing and also when operating on ground.**

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General description

The SMS provides situational awareness on airports to pilots through the functions that follow:

- **Take-off and Landing Awareness Function (TLAF)**
- **Optional Airport Moving Map (AMM) (POST SB700-35-7503).**

The TLAF gives visual and aural alerts to help the pilots ensure that the aircraft can take off or land safely. The TLAF is performed by the Flight Management System (FMS). The optional AMM provides a top-down graphical representation of airport map data integrated into the standard navigation map to assist the flight crew in orientation on the airport surface.

The SMS also provides visual indications of FMS target runways on Jeppeson airport charts. Runways are highlighted:

- With a directional arrow over takeoff and landing runways
- Magenta highlight for landing or cyan for takeoff.

Fig 1 Surface management system - Jeppesen map visual indication

The RUNWAY Push-Button Annunciator (PBA) switch, allows inhibition of SMS TLAF alerts. The PBA is connected to the Data Concentration Unit Module Cabinets (DMCs). Visual annunciations are provided in the form of text messages on the Primary Flight Display (PFD) and the Head-Up Display (HUD). Aural annunciations are issued through the cockpit-installed audio system that receives the audio signals for the alert from the Radio Interface Units (RIU), which in turn is commanded by the SMS through the Engine Indication and Crew Alerting System (EICAS).

Terrain awareness warning system control panel

The terrain awareness warning system (TAWS) control panel (34-42-00) allows the pilots to inhibit the SMS aural and CAS alerts using the RUNWAY OFF push button assembly (PBA). The SMS provides the state of the RUNWAY PBA pressed or not. Pushing the RUNWAY PBA causes OFF to illuminate as well as posting SMS RUNWAY OFF status message in the EICAS display. Pushing the RUNWAY PBA again restores the SMS associated aural and CAS messages alerts.

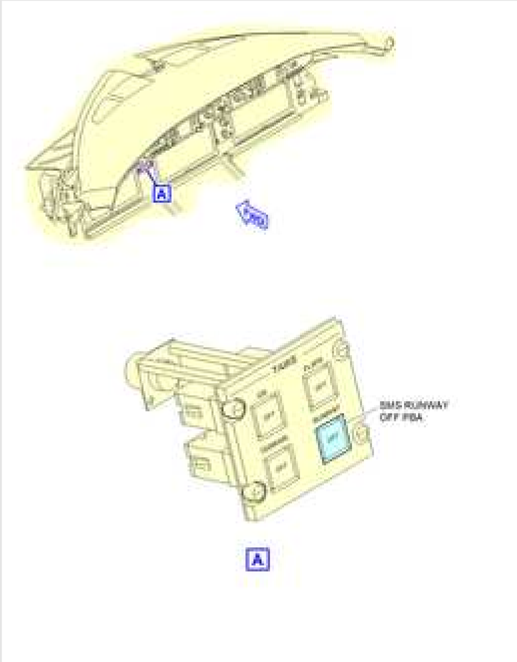

Fig 2 TAWS control panel - SMS runway OFF PBA location

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Common computing module

The SMS is a distributed system which depending upon the number of flight management system (FMS) and integrated flight information system (IFIS) installations may involve 2 CCMs (refer to Integrated processing system ATA 31-41-00) in each Integrated Processing Cabinet (IPC) as well as both DCMs in each DMC and the AFDs.

The CCM is a line-replaceable module (LRM) that is physically mounted in the IPC 1, IPC 2 and IPC 3. The CCMs receive inputs from the aircraft avionics subsystems, usually through the data concentration system (DCS). The CCMs use these inputs, in conjunction with data stored in the CCM navigation database, to create a visual display of the navigation information required to get the aircraft to its destination. The CCMs also process database information to provide aircraft performance outputs to the display units and other aircraft systems. The CCMs that hold the SMS applications are in the first slot of each IPC, along with the FMS applications.

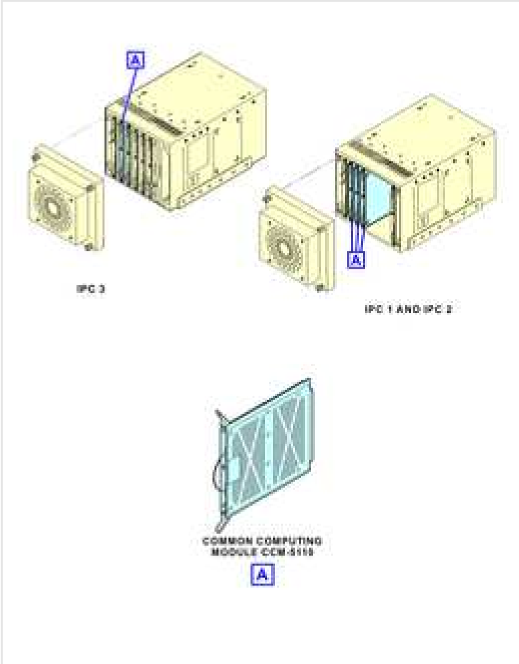

Fig 3 Common computing module - Component location

The CCM has 6 operational modes as follows:

- **Normal mode - It is dependent on the results of monitors, tests and the validity checks.**
- **Software validation mode - It checks the field loaded software when run time validity check fails or after data loading the module.**
- **Halt mode - When the internal processor still demonstrates signs of a fatal fault following 3 unsuccessful attempts to restart.**
- **Invalid strapping mode - When a discrepancy in strapping or in cabinet position is detected.**
- **Invalid platform configuration mode - When an error specific to platform configuration is detected.**
- **Data mode - When the CCM receives and acknowledges a data loading request via the data load enable discrete.**

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Aircraft moving map (Optional) (POST SB700-34-7503)

On aircraft post SB700-34-7503, the AMM provides a top-down graphical representation of airport map data integrated into the standard navigation display, which in turn provides both a heading-up and a north-up orientation that improves situational awareness with regards to take-off/landing and ground operations.

The AMM also provides the following functions:

- **Range availability**
- **Ranges below 2 NM with a minimum of 1/8 NM**
- **Textual airport information**
- **Airport information block**
- **Highlighting based on FMS flight plan**
- **Graphical runway identifier information**
- **Runway identifier symbols.**

The AMM requires an application software which is located on the CCM module installed in the left IPC. The AMM is enabled by an encrypted application key.

The application software generates a detailed airport image from the ARINC 816 airport maps database only if the airport image is in the database. The airport image information in the ARINC 816 includes runways, taxiways, apron features, buildings and water bodies.
The application software also uses ARINC 424 airport runways database for the airport that is not present in the ARINC 816. The AMMMA-6000 displays the airports in low detail, presenting runway surfaces and the markings.

In addition, both ARINC 816 and ARINC 424 provides an own-ship symbol, runway highlight and runway designation information.

The system uses GPS data and attitude/heading inputs from the Inertial Reference System (IRS) to produce a 2D navigation solution to support positioning of an own-ship symbol on the airport map by the AFD. An estimated position error is output to the flight displays such that an error circle representative of estimated position error may be displayed about the aircraft symbol. If the accuracy of the data is not sufficient by the application software, the own-ship symbol is removed from the AMM display.

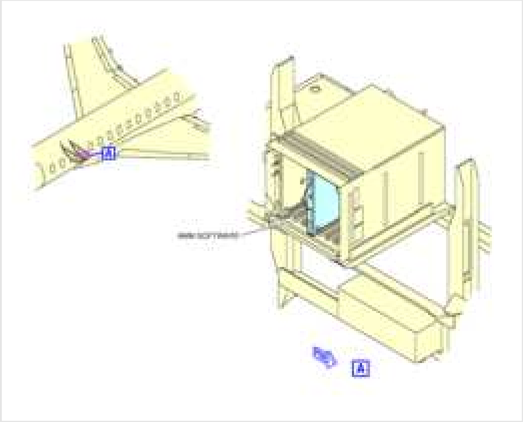

Fig 4 AMM application software/CCM (optional) - Component location

System operation

During normal operation, the SMS works in conjunction with the FMS, and displays overlays on IFIS runway charts in conjunction with the EFIS display system. The map displays aircraft moving map (AMM) on a multi function window (MFW) when selection made from symbols menu. SMS alert modes use existing airport data and FMS takeoff and landing performance data. SMS algorithms are a component of the FMS.

The SMS uses the following types of data:

- GPS
- Throttle position
- Heading
- Flight plan
- Aural inhibit

- runway highlight ON/OFF
- Encrypted key

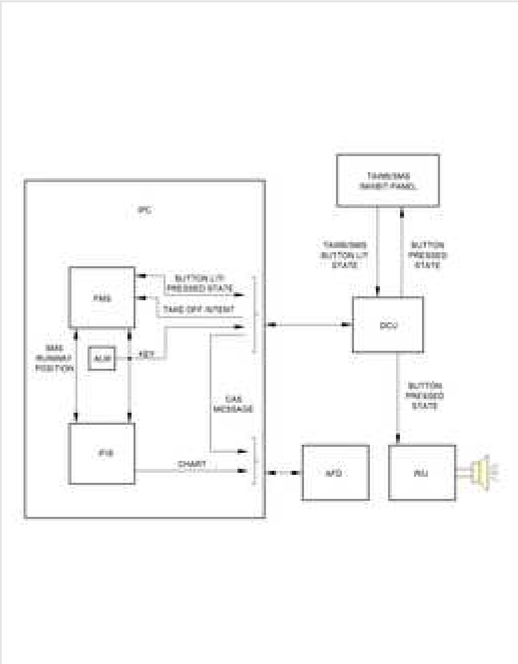

Fig 5 Surface management system - Block diagram

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Control and display

The TAWS control panel RUNWAY OFF PBA

The TAWS control panel allows the pilots to inhibit the SMS aural and CAS alerts using the RUNWAY OFF PBA. The SMS provides the state of the RUNWAY PBA pressed or not. Pushing the RUNWAY PBA causes OFF to illuminate as well as posting SMS RUNWAY OFF status message in the EICAS display. Pushing the RUNWAY PBA again restores the SMS associated aural and CAS messages alerts.

Indication

The SMS supplies the following data:

- Aural advisories
- Highlighted runways
- CAS message
- SMS status
- SMS health

SMS alerts

The alerts are only for abnormal situations. There are five alert modes for takeoff and landing. Each alert consists of an aural message and a PFD flag.

The five alerts of the SMS are as follows:

Table 1 - SMS aural and visual alerts

Mode	Condition	Aural alert	PFD message	HUD message	Severity level
1	Takeoff not on a runway	'Not a Runway, Not a Runway'	RUNWAY (Red)	RUNWAY (Boxed)	Warning
2	Takeoff runway too short	'Short Runway, Short Runway'	RUNWAY (Red)	RUNWAY (Boxed)	Warning
3	Takeoff runway disagrees with FMS departure runway	'Runway Miscompare, Runway Miscompare'	RUNWAY (Amber)	RUNWAY (Un-boxed)	Caution
4	Landing not on runway	'Not A Runway, Not A Runway'	RUNWAY (Red)	RUNWAY (Boxed)	Warning
5	Landing runway too short	'Short Runway, Short Runway'	RUNWAY (Red)	RUNWAY (Boxed)	Warning

SMS alert messages flash according to the standard flash routine with PFD messages for Modes 1, 2, 4 and 5 displayed in red and PFD messages for Mode 3 displayed in amber.

On the HUD, Warning messages are boxed and Caution messages are un-boxed. In the event of conflict, TAWS messages will always have priority over SMS alert messages.

The radio interface unit (RIU) receives an SMS advisory mode, takes the aircraft aural prioritization scheme into account, and plays the appropriate aural. Audio is controlled using the audio control panel (ACP). The default state of the SMS is ON, so that it does supply PFD flag and aural alerts.

Under circumstances where more than one conditions are met, only the SMS mode with the highest priority is activated. During either starting takeoff or before landing, the order of priority is ranked in the following:

- Mode 1 or Mode 4: Not a Runway
- Mode 2 or Mode 5: Short Runway
- Mode 3: Runway Miscompare

Note:
At take off speed V1 SMS modes 1-3 are inhibited.

SMS alert mode 1 - Takeoff not on a runway

The SMS compares the current aircraft position and heading with all the runways at the departure airport.

The SMS then supplies this alert if:

- The aircraft is not aligned with any runway
- Takeoff thrust is applied or
- Ground speed >= 35 knots and
- Ground speed < V1

The IRS heading is used for this comparison below 35 Knots. The GPS is used at or above 35 Knots.

Audio flight deck effect:

- Not a runway, not a runway

PFD / HUD annunciation:

- RUNWAY in red on PFD
- Boxed monochrome RUNWAY on HUD

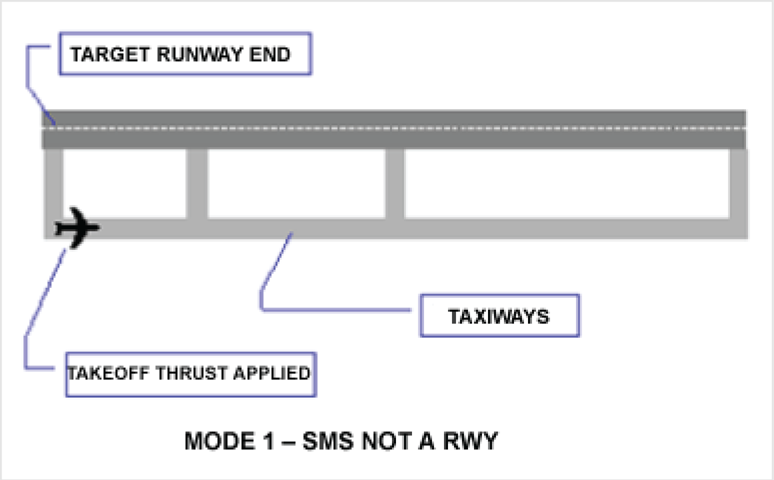

Fig 6 Mode 1 - SMS not a runway

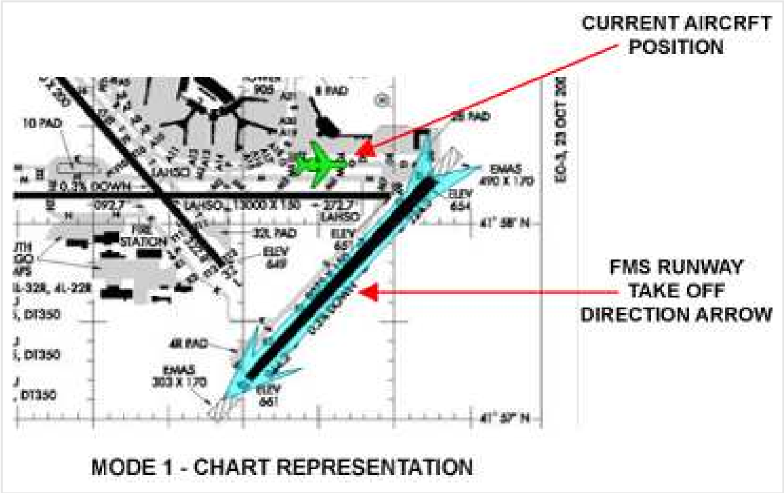

Fig 7 Mode 1 - Chart representation

SMS alert mode 2 – Takeoff runway too short

The SMS compares the required takeoff length computed by the FMS to the actual runway distance remaining, based on the current aircraft position and the end of the runway.

The SMS then supplies this alert if:

- The runway distance remaining is insufficient for takeoff
- Takeoff thrust is applied or
- Ground speed >= 35 knots and
- Ground speed < V1

The IRS heading is used for this comparison below 35 Knots. The GPS is used at or above 35 Knots.

Audio flight deck effect:

- Short runway, short runway

PFD / HUD annunciation:

- RUNWAY in red on PFD
- Boxed monochrome RUNWAY on HUD

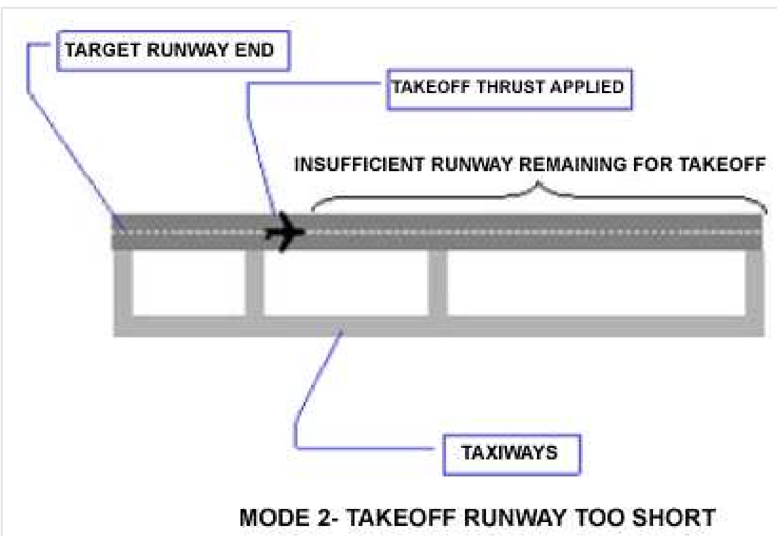

Fig 8 Mode 2 - Takeoff runway too short

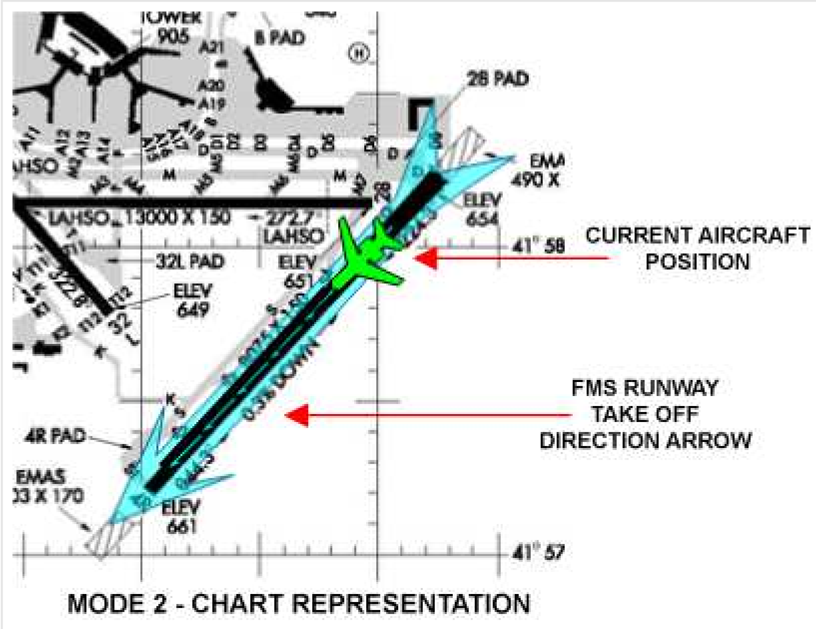

Fig 9 Mode 2 - Chart representation

SMS alert mode 3 – Takeoff runway disagrees with FMS departure runway

The SMS compares the actual takeoff runway to the FMS (Flight Plan) departure runway, based on the aircraft’s current position and heading.

The SMS then supplies this alert if:

- The aircraft is on a runway different than the flight plan departure runway
- Takeoff thrust is applied or
- Ground speed >= 35 knots and
- Ground speed < V1

The IRS heading is used for this comparison below 35 Knots. The GPS is used at or above 35 Knots.

Audio flight deck effect:

- Runway miscompare, runway miscompare

PFD / HUD annunciation:

- RUNWAY in yellow on PFD
- Monochrome RUNWAY on HUD

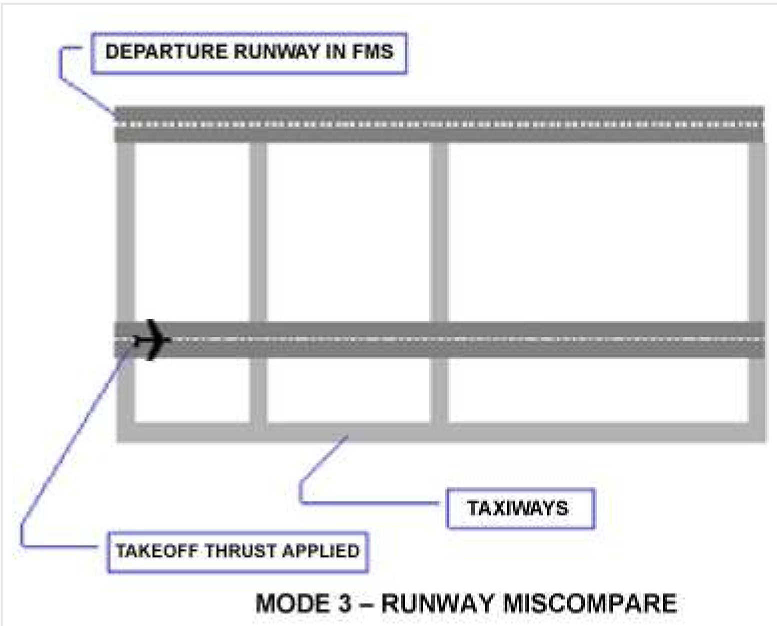

Fig 10 Mode 3 - Takeoff runway disagrees with FMS departure runway

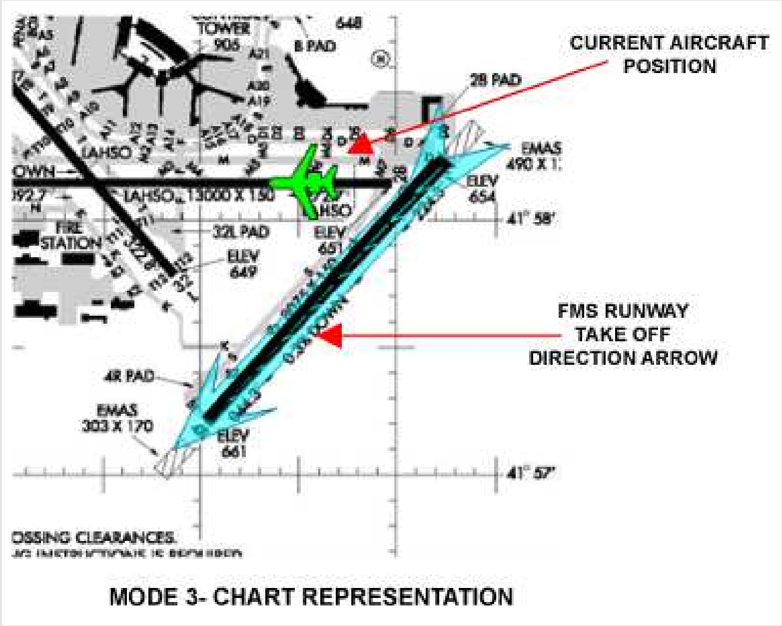

Fig 11 Mode 3 - Chart representation

SMS alert mode 4 – Landing not on runway

The SMS compares the aircraft’s heading and position to the runways of the nearest airport.

The SMS then supplies this alert if:

- The aircraft is not aligned with any runway
- Gear is down and locked
- The aircraft is **200** or fewer feet above the highest runway at the airport

The aircraft is considered aligned with a runway if the aircraft is inside a trapezoidal zone before the start of the runway, and the aircraft's heading intersects the runway.

Audio flight deck effect:

- Not a runway, not a runway

PFD / HUD annunciation:

- RUNWAY in red on PFD
- Boxed monochrome RUNWAY on HUD

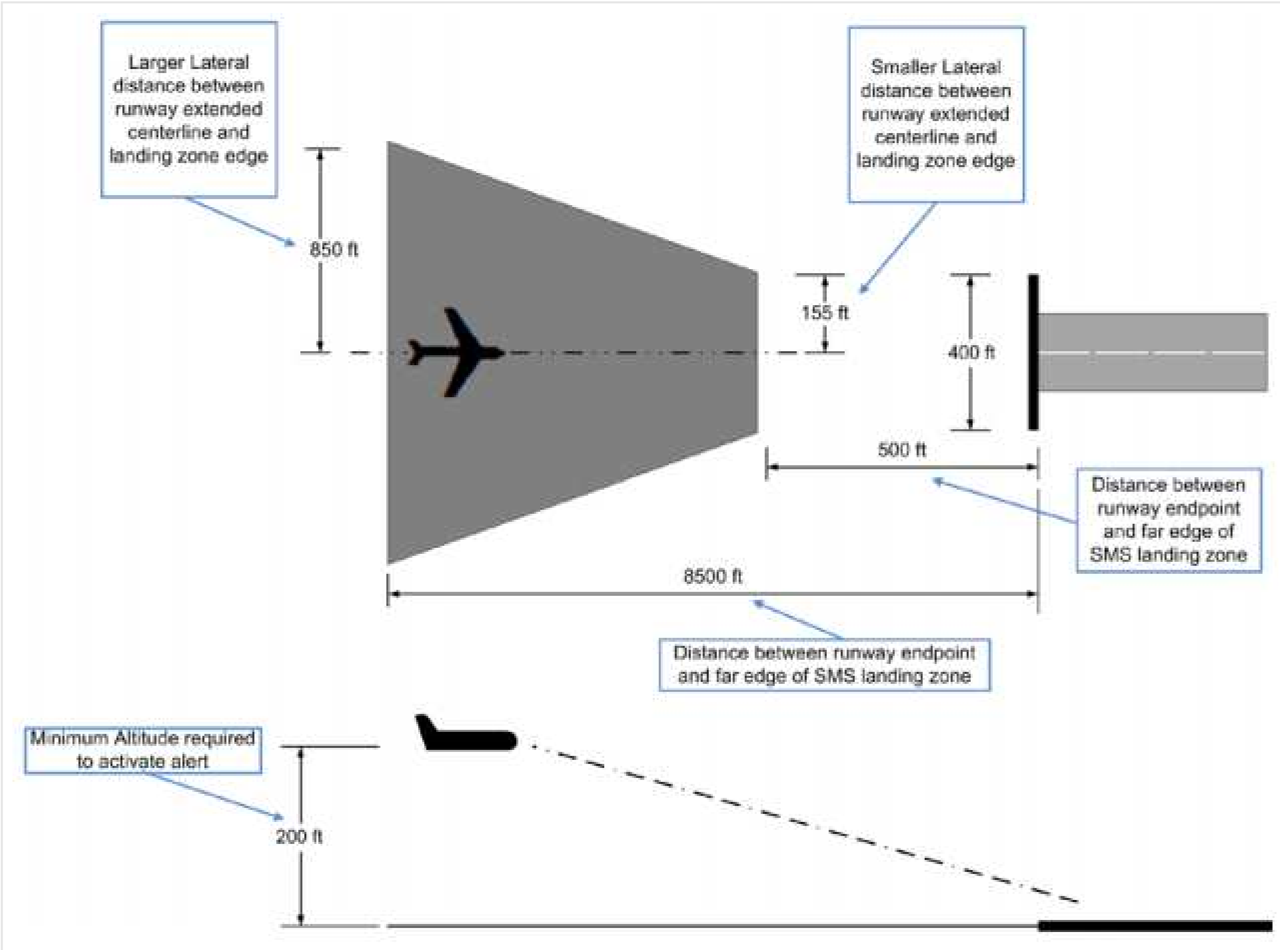

Fig 12 Mode 4 - Landing not on runway

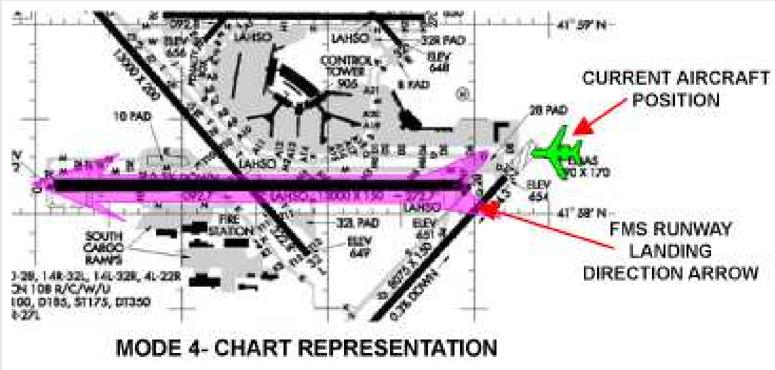

Fig 13 Mode 4 - Chart representation

SMS alert mode 5 – Landing runway too short

The SMS compares the length of the aligned landing runway to the required landing length as computed by the FMS.

The SMS then supplies this alert if:

- The aircraft is aligned with a runway that is too short
- Gear is down and locked
- The aircraft is 200 or fewer feet above the highest runway at the airport

The runway is considered short based on aircraft performance parameters and runway length.

Audio flight deck effect:

- Short runway, short runway

PFD / HUD annunciation:

- RUNWAY in red on PFD
- Boxed monochrome RUNWAY on HUD

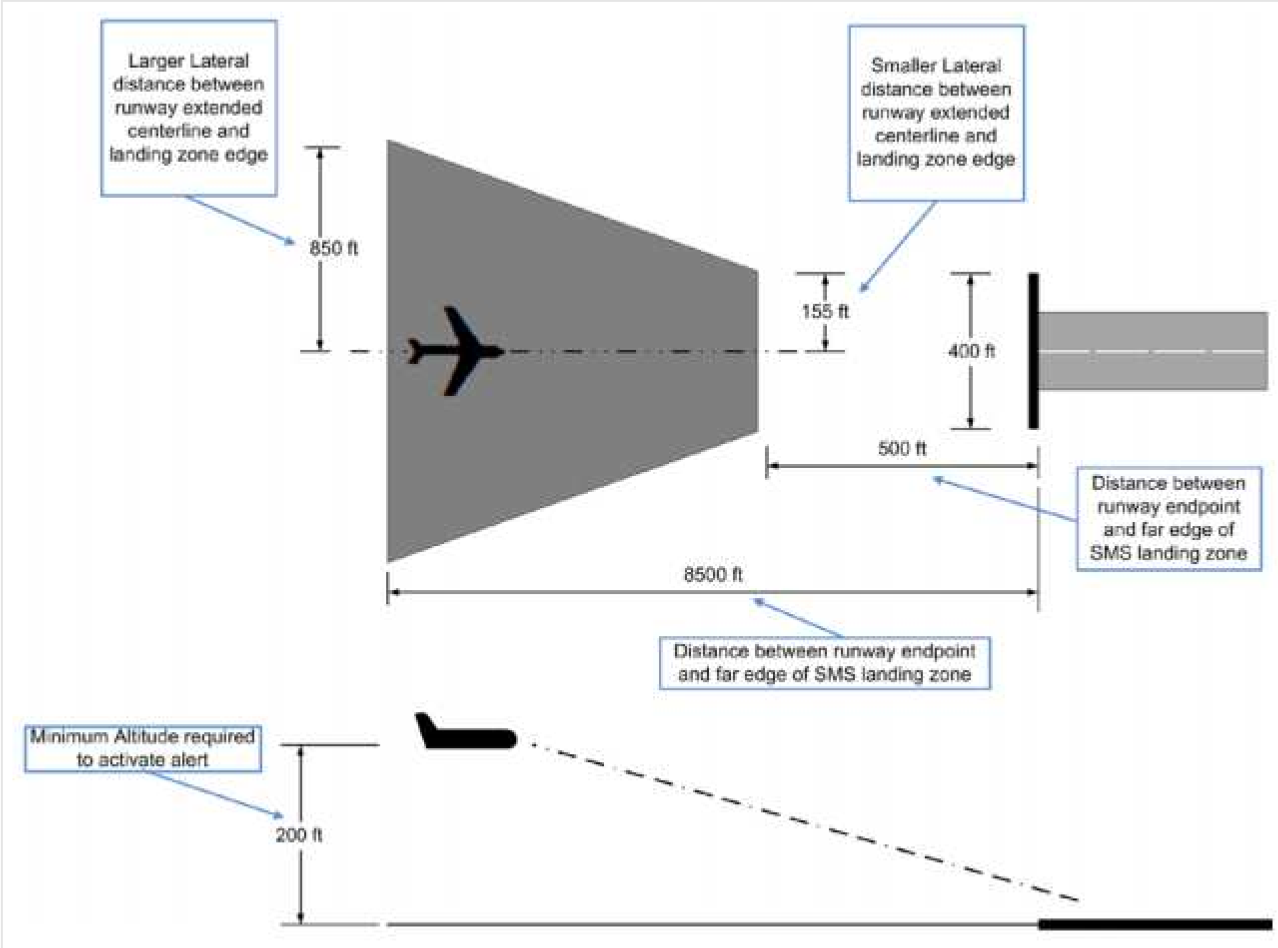

Fig 14 Mode 5 - Landing runway too short

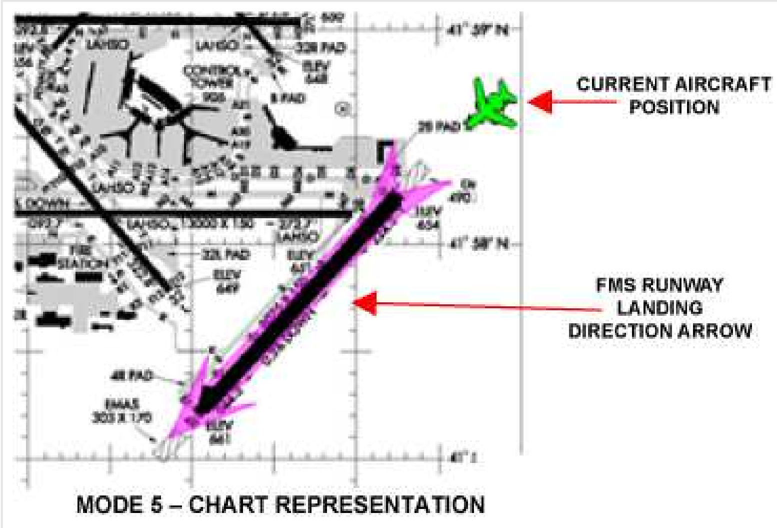

Fig 15 Mode 5 - Chart representation

AMM SMS alerts

The AMM fault and status messages that display on the airport map message field are as below:

Table 2 - AMM fault messages

Message	Type	Description
LOADING AIRPORT	WHITE	When processing of an airport map is incomplete.
OFF AIRPORT MAP	WHITE	When airport is not in view.
MAP POSITION FAULT	CYAN	When present position of map or heading is invalid or missing.
ARINC MAP FAULT	CYAN	When AMM appliation is not capable of processing ARINC 424 and ARINC 816 airport mapping data and when the map range is less than 2 NM.

System interface

The SMS is a distributed system which, depending upon the number of FMS and IFIS installations, may involve 2 CCMs in each IPC as well as both DCMs in each DMC and the AFDs.
The CCMs are LRM that are physically mounted in the IPC. The CCMs receive inputs from the aircraft avionics subsystems, usually through the data concentration system (DCS). The CCMs use these inputs, in conjunction with data stored in the CCM navigation database, to create a visual display of the navigation information required to get the aircraft to its destination. The CCMs also process database information to provide aircraft performance outputs to the display units and other aircraft systems. The CCMs that hold the SMS applications are in the first slot of each IPC, along with the FMS applications.

The SMS interfaces with the aircraft systems that follow:

- Flight management system (FMS)
- Electronic flight instrument system (EFIS)
- data concentration system (DCS)
- Global positioning system (GPS)
- Inertial reference system (IRS)

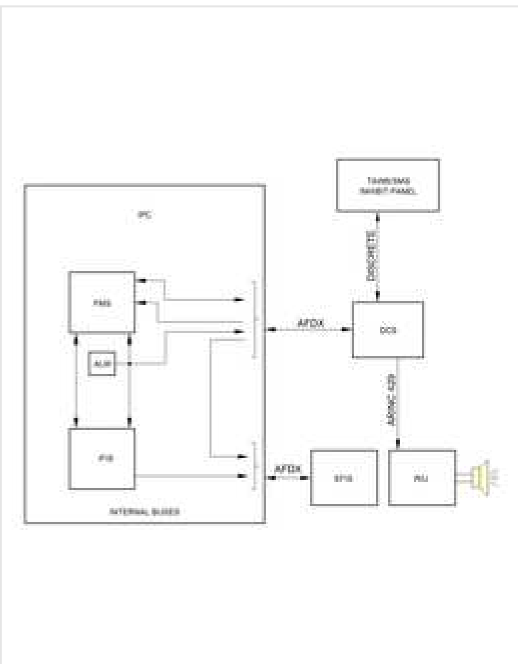

Fig 16 Surface management system - Interface block diagram

Table 3 - SMS Interface

Interfacing component	Interface description	I/O	Interface type
FMS	Performs SMS Aural Advisory Algorithm Function. Outputs SMS Advisory Mode. Manages aural inhibit button. Sends SMS Airport/Runway parameters to IFIS. Outputs SMS status.	I/O	Internal bus
EFIS	Displays CAS messages. Gets symbols from IFIS for runway charts, including highlighting if applicable. uses IFIS active airport/runway information to highlight the flight plan runway on the airport chart.	I/O	A664
DCS	Forwards advisory indications to the RIU. Checks the current SMS Advisory Mode and SMS status to see if a PFD message is necessary. Provides takeoff intent to FMS. Throttle levers angle in takeoff position	I/O	A664
GPS	Heading	I	A664
IRS	Heading	I	A664

System test

Each CCM performs a series of power-up built in test (PBIT) and continuous built in test (CBIT) to verify functionality. If any of these tests fail, an SMS fault flag is shown on the electronic flight displays.

System monitoring

SMS CAS message type

Advisory message

- SMS NOT AVAILABLE - SMS not available due to degraded position accuracy, loss of required inputs or all SMS loss
- SMS SHORT RWY INOP - Take off field length or landing field length not calculated

Status message

- SMS RUNWAY OFF - Function selected OFF on the TAWS panel

Component location

Table 4 - Component location index

Ident	Description	Location/Zone	Maintenance	Parts	Wiring

Ident	Description	Location/Zone	Maintenance	Parts	Wiring
PL35	TAWS control panel	FS230L	AMP34-42-03	IPD34-42-01	AWM34-42-00
-	CCM-5110	-	AMP31-41-13	IPD31-41-13	AWM34-42-00